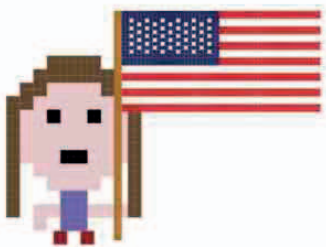


Symbols and codes

Computers use binary code to translate numbers into electrical signals. But how would a computer use binary code to store the words and characters on this page?

ASCII

The first computers each stored characters in their own unique way. This worked fine until data needed to be moved between computers. At this point, a common system was chosen, called the American Standard Code for Information Interchange (ASCII, pronounced “askey”).



▷ ASCII table

In ASCII, a decimal number value is given to each character in the upper- and lower case alphabets. Numbers are also assigned to punctuation and other characters, such as a space.

▷ ASCII in binary

Because each character has a number, that number then needs to be converted to binary to be stored in a computer.

R = 82 = 1010010

r = 114 = 1110010

▽ ASCII in Python

You can convert between ASCII and binary code in most languages, including Python.

This command prints the character, the ASCII value, and the binary value for each letter in the name “Sam”

```
>>> name = 'Sam'
>>> for c in name:
    print(c, ord(c), bin(ord(c)))
```

```
S 83 0b1010011
a 97 0b1100001
m 109 0b1101101
```

Here are the results. The beginning of each binary number is marked “0b”

SEE ALSO

◀ **180–181** Inside a computer

◀ **182–183** Binary and bases

ASCII			
32	SPACE	64	@
33	!	65	A
34	"	66	B
35	#	67	C
36	\$	68	D
37	%	69	E
38	&	70	F
39	'	71	G
40	(	72	H
41	)	73	I
42	*	74	J
43	+	75	K
44	,	76	L
45	-	77	M
46	.	78	N
47	/	79	O
48	0	80	P
49	1	81	Q
50	2	82	R
51	3	83	S
52	4	84	T
53	5	85	U
54	6	86	V
55	7	87	W
56	8	88	X
57	9	89	Y
58	:	90	Z
59	;	91	[
60	<	92	\
61	=	93	]
62	>	94	^
63	?	95	_
96	`	127	DELETE
97	a		
98	b		
99	c		
100	d		
101	e		
102	f		
103	g		
104	h		
105	i		
106	j		
107	k		
108	l		
109	m		
110	n		
111	o		
112	p		
113	q		
114	r		
115	s		
116	t		
117	u		
118	v		
119	w		
120	x		
121	y		
122	z		
123	{		
124			
125	}		
126	~		